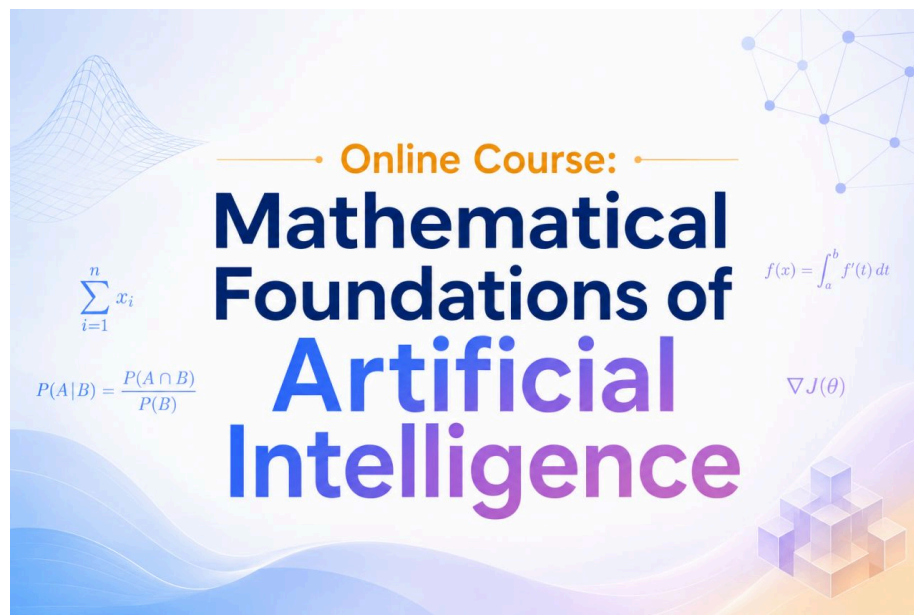


**Mathematical Foundations
of Artificial Intelligence**
A Universal Framework for Understanding Modern AI
Module descriptions and Course overview
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Introduction and Motivation

Mathematical Foundations of Artificial Intelligence is an online course that allows you to learn AI from first principles - in a unique and simple way. All you need is the basic maths you need to GCSE/A levels i.e. Linear Algebra, Statistics, Probability theory and Optimization.

Places are limited since this is an experiment. It also has a limited promotional pricing.

This course makes an ambitious claim: It aims to future proof your understanding of AI - because you are understanding AI from first principles

Admittedly, this approach is not for everyone - but based on my teaching - you will be surprised how many people like it even if they are apprehensive in the beginning!

Here's why

1. It's based on things you already know ex Linear algebra etc at some point in your life. We just put it together in a way which relates the fundamentals to ALL of AI
2. Learning only a few foundations means you can focus on how things apply
3. A lot fewer equations - yes you have equations and maths - but its applied maths. Its not an endless sea of cryptic equations which bamboozle you
4. We focus on modern AI - Deep neural networks and beyond. This helps you to apply it to real problems
5. AI changes very rapidly, this is a very good way to understand the foundations so that you build your learning from first principles.
6. The most important reason is - you learn best when you understand how AI works 'under the hood' and trust me - its not that hard as it sounds.

This course is for Developers, Researchers, Product managers, Engineers and anyone who is curious

Course format:

Online. 4 months. Self paced. Materials available beyond course. Limited places Office hours. Scenario based quiz where you can use LLMs. A hall of fame opportunity for more advanced tests.

Notes:

- 1) We encourage mathematical thinking
- 2) We use a problem first approach with a spiral learning pedagogy (we repeat concepts over the course in increasing levels of complexity)
- 3) The course has a comprehensive Glossary and other supporting materials

Summary of Modules

Module 1 – A New Way to Think About AI

This opening module introduces the central philosophy of the course: understanding artificial intelligence through a small number of enduring mathematical ideas rather than a catalogue of algorithms. Students are introduced to the five recurring lenses—Learning, Representation, Computation, Optimisation, and Theory—which together provide a universal framework for analysing any AI system, from classical machine learning models to modern foundation models.

The module also examines the paradox of deep learning, the distinction between statistical and algorithmic modelling, and the mathematical foundations required for the rest of the course, establishing a way of thinking that remains relevant as AI continues to evolve.

Module 2 – Learning: Formulating the Problem

Every AI system begins by defining a learning problem, and this module explores the mathematics behind that process. Students learn how data, probability distributions, hypothesis spaces and inductive bias determine what a model can and cannot learn. The module develops a unified understanding of supervised, unsupervised, self-supervised and reinforcement learning, while introducing concepts such as generalisation, function approximation and the No Free Lunch theorem. By the end, learners understand that successful AI depends as much on formulating the problem correctly as it does on selecting an algorithm.

Module 3 – Representation: How AI Builds Abstractions

Modern AI succeeds because it learns useful internal representations rather than relying on manually engineered features. This module examines how neural networks transform raw data into increasingly abstract representations, connecting ideas from geometry, information theory and representation learning. Students explore embeddings, latent spaces, manifold learning, information bottlenecks, transfer learning and mechanistic interpretability, developing an intuition for how deep learning captures semantic structure and why learned representations underpin the remarkable capabilities of modern AI systems.

Module 4 – Computation: How Intelligence Is Implemented

This module examines the computational machinery that makes modern AI possible. Beginning with the basic neuron and progressing through multilayer perceptrons, convolutional neural networks and transformers, students learn how neural architectures perform computation and why different architectures excel on different tasks. Particular emphasis is placed on the mathematics of attention, transformer blocks, positional encoding, residual connections and feed-forward networks, showing how a relatively small set of computational building blocks has become the foundation of today's most powerful AI models.

Module 5 – Optimisation: Discovering Useful Solutions

Learning is fundamentally an optimisation problem, and this module develops the mathematical tools that enable AI systems to discover effective solutions. Students study loss functions, gradient descent, stochastic optimisation, momentum, Adam, learning-rate schedules and regularisation, while exploring the geometry of high-dimensional loss landscapes. Classical concepts such as Ridge, Lasso and the bias–variance trade-off are connected to modern deep learning, culminating in an explanation of why simple optimisation algorithms are remarkably successful even when training extremely large neural networks.

Module 6 – Theory: Why AI Works

Although modern AI systems achieve extraordinary performance, a complete theoretical explanation remains an open scientific challenge. This module surveys the leading mathematical theories that attempt to explain deep learning, including classical statistical learning theory, double descent, benign overfitting, scaling laws, the manifold hypothesis, information theory and mechanistic interpretability. Rather than presenting a single definitive answer, the module equips learners to critically evaluate competing explanations and understand both the strengths and limitations of current AI theory.

Module 7 – Mathematical Models Across AI

This module traces the mathematical evolution of artificial intelligence from classical statistical models to today's generative foundation models. Beginning with linear regression, generalized linear models and support vector machines, the course progresses through multilayer perceptrons, autoencoders, variational autoencoders, GANs and diffusion models. Rather than studying these models in isolation, students discover the mathematical principles that connect them, revealing a continuous progression in which increasingly expressive representations, optimisation methods and inductive biases give rise to more capable AI systems.

Module 8 – Frontiers of Artificial Intelligence

The final module explores the active research frontiers shaping the future of AI. Students investigate topics such as mechanistic interpretability, scaling laws, world models, scientific machine learning, symbolic AI, foundation models, multimodal learning and agentic AI. Throughout, the five mathematical lenses introduced at the beginning of the course are applied to emerging research, enabling learners to analyse new developments systematically while appreciating the many open questions that continue to drive progress in artificial intelligence.

Learning Objectives for the course

Module 1 – A New Way to Think About AI

- Explain why modern deep neural networks generalise despite classical statistical theory predicting overfitting.
- Analyse any AI system using the five mathematical lenses.
- Explain why organising AI around mathematical ideas is more durable than organising it around algorithms.
- Compare statistical modelling and algorithmic modelling and identify where modern AI fits.
- Decompose an AI paper using the five-lens framework.
- Identify the mathematical tools required to understand modern AI.

Module 2 – Learning

- Explain learning as selecting a function from a hypothesis space.
- Formulate a machine learning problem mathematically from data and distributions.
- Distinguish supervised, unsupervised, self-supervised and reinforcement learning.
- Select the appropriate mathematical formulation for a given AI task.
- Explain different forms of generalisation and why they matter.
- Describe every AI model as a function approximator.
- Analyse how hypothesis spaces determine what a model can learn.
- Explain how architecture imposes inductive bias.

- Justify why inductive bias is essential for successful learning.
- Use the No Free Lunch theorem to explain why unrestricted learning is impossible.
- Explain how reinforcement learning differs fundamentally from supervised learning.

Module 3 – Representation

- Explain why learned features replaced expert-designed rules.
- Describe how AI automatically discovers useful representations.
- Interpret neural representations as geometric structures.
- Evaluate the quality of learned representations.
- Distinguish features from representations.
- Explain hierarchical abstraction in deep learning.
- Explain how the information bottleneck balances relevance and compression.
- Evaluate the relationship between compression and intelligence.
- Explain why embeddings capture semantic similarity.
- Explain why knowledge transfers between tasks.
- Explain how multiple modalities can share a representation space.
- Explain the manifold hypothesis for language.
- Analyse the geometry of latent spaces.
- Explain why neural network functions are implicit rather than explicit.
- Interpret trained neural networks as executable computational procedures.
- Explain approaches for reverse engineering neural networks.
- Explain why natural data lies on low-dimensional manifolds.
- Synthesize the geometric principles underlying modern AI.

Module 4 – Computation

- Explain how neural networks compute functions.
- Evaluate how architectural choices affect learning.
- Explain the strengths and weaknesses of MLPs.
- Explain why convolutions are effective for images.
- Explain the computational advantages of transformers.
- Explain how parameters encode learned knowledge.
- Relate convolutional inductive bias to image statistics.
- Explain why attention enables flexible computation.
- Describe mathematically how attention computes relationships.
- Explain why multiple attention heads improve learning.
- Describe the computational role of transformer blocks.
- Explain why positional information is required.
- Explain how residual connections enable deep learning.
- Describe the computational role of feed-forward networks.
- Trace the complete computation performed by a transformer.
- Evaluate competing explanations for transformer success.
- Explain why transformers generalise across modalities.

Module 5 – Optimisation

- Explain optimisation as the mathematical engine of learning.
- Describe the complete training pipeline.
- Select appropriate loss functions for different learning problems.
- Interpret optimisation geometrically.
- Explain how gradient descent finds useful solutions
- Explain why stochasticity improves optimisation.
- Explain how momentum accelerates convergence.

- Compare Adam with SGD and Momentum.
- Explain the role of learning-rate schedules.
- Explain how regularisation improves generalisation.
- Explain L2 regularisation mathematically.
- Explain why L1 regularisation performs feature selection.
- Compare Ridge, Lasso and Elastic Net.
- Analyse the bias–variance trade-off.
- Explain regularisation using Bayesian priors.
- Evaluate the Lottery Ticket Hypothesis.
- Explain why optimisation succeeds in deep learning.

Module 6 – Theory

- Explain the foundations of classical statistical learning theory.
- Explain the theoretical prediction of overfitting.
- Explain the double-descent phenomenon.
- Explain why perfect fitting can still generalise.
- Explain the theoretical role of manifolds.
- Explain empirical scaling laws.
- Analyse the phenomenon of emergence.
- Critically evaluate modern evidence for Lottery Tickets.
- Explain interpretability as a scientific theory.
- Evaluate compression-based theories of intelligence.
- Explain entropy, cross-entropy and KL divergence in learning.
- Critically discuss the major open questions in AI theory.

Module 7 – Mathematical Models

For every model: Analyse the mathematical assumptions, computational mechanism, optimisation process, and strengths and limitations of the model.

- Linear Regression – Explain the mathematical assumptions underlying linear regression.
- GLMs – Explain how GLMs unify many statistical models.
- SVMs – Explain why kernels extend linear models.
- Generative Models – Compare discriminative and generative learning.
- Diffusion Models – Explain iterative denoising mathematically.
- Mathematics Unifying Generative AI – Compare VAEs, GANs and diffusion through transport maps.
- Neural vs Non-neural Models – Evaluate when neural and non-neural models are appropriate.

Module 8 – Frontiers

- Evaluate current interpretability research.
- Evaluate scaling-law research.
- Evaluate world models.
- Evaluate scientific machine learning.
- Compare symbolic and neural AI.
- Explain why tree ensembles remain competitive.
- Evaluate foundation models.
- Evaluate multimodal learning.
- Analyse agentic AI.
- Compare neural and non-neural paradigms.
- Critically discuss major open research questions.

Three anchoring questions

I like to express to open through three questions

1. Is there a mathematical way to define intelligence?
2. If current LLM models are simply predicting next sequences (sentences) - why are they so effective especially across domains? And by extension, what's the connection between sentence predication and writing code?
3. What does the success of reasoning loops tell us about intelligence?

Module 1 – A New Way to Think About AI

Theme *Understanding AI through recurring mathematical ideas rather than algorithms.*

- Introduces the five recurring mathematical lenses — Learning, Representation, Computation, Optimisation, Theory — that the rest of the course uses to read every modern AI system, from linear regression to large language models.
- The goal is to equip learners with a framework that outlasts any specific architecture - current and future.

1.1 The Paradox of Modern AI

- classical statistical learning theory predicted that vastly over-parameterised models would overfit catastrophically, yet modern deep learning systems with billions of parameters generalise extraordinarily well.
- The paradox is not that the mathematics has failed but that existing theory is incomplete, and resolving it is the defining scientific problem of contemporary machine learning.

1.2 Five Questions That Explain Modern AI

- Introduces the five interlocking questions that organise the course — what is being learned, how is it represented, how is it computed, how are good solutions discovered, and why does any of it work.
- Together these become a reusable lens for approaching any modern AI paper or architecture, not just those that exist today.

1.3 The Architecture of the Course

- Explains why the course is organised around five lenses rather than around algorithms or applications.
- The aim is to teach learners to think rather than memorise, so that when a new architecture appears, they can analyse it by asking the same five questions instead of reaching for a fresh catalogue of techniques.

1.4 The Two Cultures of AI

- Previews Leo Breiman's distinction between the data modelling culture, which assumes a known stochastic generative process and seeks interpretable parameter estimates, and the algorithmic modelling culture, which treats the data-generating process as a black box and seeks predictive accuracy.
- Modern deep learning is positioned as a fusion of both — algorithmic in its refusal to impose parametric assumptions, statistical in its loss functions and probabilistic training objectives.

1.5 Mathematical Prerequisites

- Surveys the mathematics learners need: linear algebra (vectors, matrices, eigendecomposition), calculus (gradients, the chain rule), probability (distributions, conditioning, MLE) and optimisation (gradient descent, convexity).
- The course assumes only the basics taught at GCSE/A-level and develops everything else as it goes.

Module 2 – Learning: Formulating the Problem

Theme *Every AI system begins by defining the problem correctly.*

Module 2 develops the first lens. Before any architecture or optimiser can be chosen, the learning problem itself must be made precise — what is being approximated, from what data, under what assumptions about how the data was generated, and under what constraints that make generalisation possible. The module culminates in the argument that without inductive bias, learning is impossible.

2.1 Functions, Approximation and Learning

- Study each model as a mathematical function — linear regression, polynomial regression, deep networks — as a parameterised function, and training as function selection from the hypothesis space defined by the architecture.
- The chain is made precise: observe data, choose an architecture, define a loss, search the hypothesis space with gradient descent, and select a particular function that (one hopes) generalises.

2.2 Data, Distributions and Learning Problems

- States the supervised learning problem as a finite sample of input–output pairs is drawn i.i.d. from an unknown distribution, a hypothesis space is chosen, a loss is minimised on the training set, and the central concern is whether the resulting function will perform well on new data.
- The submodule formalises the generalisation gap and flags the increasingly important distinction between in-distribution and out-of-distribution performance.

2.3 Types of Learning

- Distinguishes the four canonical paradigms — supervised, unsupervised, self-supervised and reinforcement learning — by what signal supervises the learner.
- Self-supervised learning, in particular, is highlighted as the engine behind modern foundation models, where structure in the data itself supplies the supervision.

2.4 The Landscape of AI Tasks

- Establishes the taxonomy of tasks the course moves between: classification, regression, generation, sequence modelling and representation learning.
- Each task is paired with its standard loss function and archetypal applications, providing a clean reference for later chapters.

2.5 Generalisation

- Refines the colloquial notion of generalisation into a hierarchy: classical same-distribution generalisation, transfer to related tasks, few-shot learning, zero-shot generalisation from natural-language instructions, and compositional generalisation to novel combinations.
- The depth of generalisation depends on the abstraction level of what the model has learned.

2.6 Function Approximation

- Casts every machine learning model as a function approximator with an input space, an output space, and a parameterised family of functions between them.
- We separate the function classes (determined by architecture) from the learned function (determined by data, loss, optimisation and initialisation) and use this to clarify expressiveness, function selection and inductive bias.

2.7 Hypothesis Spaces

- Formalises the hypothesis space H as the set of all functions a given architecture can represent as its parameters vary, illustrated for linear regression, polynomial regression, ReLU networks and transformers.
- Addresses the deeper question i.e. is not whether a function is representable in principle, but whether it can be represented efficiently, learned reliably and generalised under realistic constraints.

2.8 Architecture and Hypothesis Space

- Explores architecture as a set of structural assumptions about the data-generating process. Convolution encodes locality and translation equivariance; recurrence assumes sequential ordering; attention assumes content-dependent relational structure.
- The design question is which architectural biases align with the statistical structure of the problem.

2.9 Inductive Bias

- Defines inductive bias as the set of assumptions a learning algorithm uses to choose among hypotheses that are equally consistent with the training data.
- Bias is not a defect but a precondition for generalisation, and its sources include architecture, optimisation, regularisation, initialisation and computational constraints.

2.10 Why Learning Is Impossible Without Inductive Bias

- Makes the formal case that finite data alone cannot determine a unique extension to unseen inputs — for any consistent fit, many other consistent extensions exist that disagree elsewhere.
- The No Free Lunch theorems show that, averaged uniformly over all possible target functions, all learning algorithms perform identically, so inductive bias is precisely what supplies the necessary restrictions.

2.11 Reinforcement Learning as a Different Learning Paradigm

- Introduces reinforcement learning as a fundamentally different formulation: an agent interacts with an environment, takes actions, observes consequences, and receives delayed rewards.
- Unlike supervised learning, the learner must trade off exploration against exploitation and reason about long-horizon credit assignment.

Module 3 — Representation: How AI Builds Abstractions

Theme *Modern AI succeeds because it learns representations.*

Module 3 develops the geometry-and-semantics layer. Where classical AI relied on hand-engineered features, modern systems learn their own representations as a by-product of training. The module treats representations as geometric objects — points, manifolds, latent spaces — and connects compression, transfer and interpretability through a unified geometric view.

3.1 From Rules to Features

- Explores one of AI's oldest debates by tying it to the mathematical nature of gradient descent.
- Rules are discrete, brittle and provide no gradient signal; features are continuous, distributed, compositional and similarity-preserving.
- The submodule traces the empirical collapse of expert systems and explains why the scaling failure of rules was structural rather than incidental.

3.2 Representation Learning

- Defines representation learning as the automatic discovery of transformations that make downstream tasks easier — linearly separable, smoothly varying, disentangled, or compressed in the information-bottleneck sense.
- It contrasts learned representations with the great achievements of hand-crafted feature engineering (SIFT, HOG, bag-of-words, MFCCs) and the structural limitations those approaches faced.

3.3 Geometry of Neural Representations

- Treats learned representations as geometric objects: each input becomes a point in high-dimensional space, the dataset becomes a point cloud, and its geometry — clusters, distances, intrinsic dimension, curvature, disentanglement — encodes what the network has learned.
- Each geometric property maps onto a specific generalisation behaviour, from sample efficiency to adversarial robustness.

3.4 Learned Representations

- Defines a representation formally as a parameterised function $\phi_{\theta} : X \rightarrow Z$ that maps raw inputs into an internal vector space, with the prediction pipeline decomposing into a representation followed by a simple head.
- Three criteria for a 'good' representation are introduced: downstream task performance, transferability, and structural properties such as linear separability, disentanglement, smoothness and compression.

3.5 Features versus Representations

- Draws a precise distinction often blurred in informal use between features and representation.
- A feature is a single scalar measurement extracted from the input; a representation is a coordinated vector of features jointly optimised to encode the structure of the data.
- The submodule contrasts local (one-hot), dense and distributed encodings, arguing that distributed representations are exponentially more expressive because meaning resides in the pattern rather than in any individual dimension.

3.6 Building Abstractions

- Develops the abstraction hierarchy that emerges layer by layer in deep networks: early layers capture low-level statistics, middle layers capture mid-level structure, and deep layers capture task-relevant semantic content.
- Each successive layer performs a stage of progressive compression — keeping what the task requires and discarding what it does not.

3.7 Information Bottleneck

- Presents Tishby, Pereira and Bialek's information bottleneck principle as a mathematical criterion for what a representation should contain: maximise mutual information with the target Y while minimising mutual information with the input X .
- The submodule gives the formal objective, interprets the two terms as complexity and relevance, and provides an honest account of the empirical controversy that followed the proposed two-phase training dynamics.

3.8 Compression and Intelligence

- Connects compression to prediction through Solomonoff induction and algorithmic probability: a system that can describe data with a short program has, in a precise sense, discovered structure.
- Memorisation is not compression because the description is no shorter than the data; genuine compression requires modelling regularities, and is therefore a proxy for understanding.

3.9 Embeddings

- Explore the idea that the empirical evidence that learned representations encode semantic structure geometrically.
- The distributional hypothesis is operationalised by Word2Vec and GloVe; linear analogies ($\text{king} - \text{man} + \text{woman} \approx \text{queen}$) reveal that semantic factors are encoded as approximately orthogonal directions; and contextual embeddings (ELMo, BERT, GPT) extend the idea from types to tokens.

3.10 Transfer Learning

- Explains why features learned on one task help with another: tasks operating on the same data type share structure, and a sufficiently large model discovers that shared structure as a by-product of training.
- Foundation models can be framed as the logical endpoint of transfer learning — a single, broadly-trained representation that amortises the cost of learning general structure across all downstream tasks.

3.11 Multimodal Representations

- Explore representation learning across vision, audio and language, then examines the striking phenomenon of multimodal alignment. CLIP's contrastive image-text training is presented as the prototype of cross-modal alignment, and
- Explore the idea that sufficiently large models from different modalities converge toward a common geometric structure.

3.12 Language as a Manifold

- Applies the manifold hypothesis to language. The combinatorial space of token sequences is astronomically large, but coherent language is confined to a thin manifold within it, sculpted by syntactic, semantic, discourse and world-knowledge constraints.
- This view explains why large models do not overfit, why next-token prediction is so effective, and why scaling helps.

3.13 Latent Space Geometry

- Treats latent spaces as Riemannian manifolds whose geometry encodes learned structure.
- Develops tangent spaces, geodesics, metrics and curvature in machine-learning terms, explains why spherical interpolation (slerp) is a clean empirical probe of meaningful structure, and argues that good latent spaces are geometric structures whose properties determine generation quality, classification accuracy and interpolation behaviour.

3.14 Hidden Functions

- Introduces the chapter's organising metaphor: a trained neural network specifies a function whose behaviour is implicitly encoded in billions of parameters and is not directly readable from inspection.
- Unlike a linear model, where coefficients have transparent meaning, a deep network's hidden function emerges from layered nonlinear composition and is best understood as a computational procedure rather than a closed-form expression.

3.15 Neural Networks as Programs

- Develops the 'weights as code' analogy: a parameter vector, together with an architecture, specifies a computational procedure that solves a task.
- It contrasts engineered software, where humans specify control flow and data structures, with learned systems where the procedure emerges from optimisation, using AlphaFold as an example of a procedure too complex to engineer manually but discoverable through learning.

3.16 Mechanistic Interpretability

- Introduces mechanistic interpretability as the programme of reverse-engineering neural networks at the level of internal computations — not just what they do but how they do it.
- Probing, attribution and circuit analysis are presented as the tools by which the hidden function becomes, partially, readable.

3.17 Manifold Hypothesis

- States the central response to the curse of dimensionality: natural data — images, language, audio, protein structures — does not fill high-dimensional space uniformly but lies on or near a low-dimensional manifold within it.
- The effective dimensionality of the learning problem is governed by the manifold's intrinsic dimension, not the ambient dimension, and a network's job is to discover and exploit that low-dimensional structure.

3.18 Geometry Behind Modern AI

- Closes the module by tying together representation learning, embeddings, latent spaces and the manifold hypothesis into a single geometric picture.
- Modern AI is read as the discovery and exploitation of the hidden geometry of natural data, and this perspective is offered as the deepest available intuition for why deep learning works at all.

Module 4 — Computation: How Intelligence Is Implemented

Theme *Understanding the machinery that performs AI computation.*

Module 4 develops the architectural layer. Having established what is learned (Module 2) and how it is represented (Module 3), the course turns to the actual machinery — neurons, layers, convolutions, attention — that performs the computation. The transformer is treated as the current general-purpose computation engine, but the deeper goal is to expose the small set of primitives from which any modern architecture is built.

4.1 What Is a Neural Network?

- Builds a concrete mental picture of a neural network as a parameterised function composed of layers, each consisting of a linear transformation followed by a nonlinear activation.
- The submodule introduces neurons, layers, weights, biases and activation functions gently, contrasts a single-layer network with classical linear regression, and explains why stacking layers without nonlinearity collapses to a single linear map.

4.2 Architecture

- Introduces depth (number of layers) and width (neurons per layer) as the coarsest characterisation of a network's shape, and uses this to introduce the broader concept of architecture: the choice of depth, width, layer types, connectivity and parameter count is a pre-training design decision that encodes assumptions about the structure of the problem.

4.3 Multi-Layer Perceptrons

- Describes the MLP as the default, general-purpose architecture: fully connected, feedforward, with no structural assumptions about its input.
- Universal approximation makes it expressive in principle, but the absence of structural inductive bias makes it data- and compute-hungry in practice, which is why MLPs survive primarily as components within larger architectures.

4.4 Convolutional Networks

- Introduces the CNN as the architecture designed for data with spatial structure: small filters slide across an input, and the same filter is applied at every position, encoding the two powerful inductive biases of locality and translation invariance.

- The typical hierarchy — local features in early layers, mid-level parts in middle layers, high-level concepts in deep layers — is sketched and grounded in the geometry of natural images.

4.5 Transformers

- Sketches the transformer's defining mechanism — attention as a learned, input-dependent routing of information between every pair of positions — and frames its inductive bias as relational rather than spatial or sequential.
- The cost of attention's flexibility (quadratic scaling in sequence length) is signalled, with the full mathematical treatment developed in subsequent submodules.

4.6 Parameters and Learned Knowledge

- Makes the abstract notion of parameters concrete: weights multiply input signals, biases shift activations, and together they form the parameter vector that determines the network's behaviour.
- Parameters are not abstract numbers in a table — they are the model's learned knowledge, structured by architecture into layers, heads, channels and blocks. Two models with identical parameter counts but different architectures can have radically different capacities.

4.7 Why Convolution Works

- Explains the empirical success of convolution by aligning its inductive biases — locality, translation equivariance and parameter sharing — with the statistics of natural images.
- Convolution restricts the hypothesis space in exactly the way image data requires, and restriction is a feature, not a defect, when the restriction matches the world.

4.8 Why Attention Works

- Frames attention as a data-dependent generalisation of convolution. Where convolution applies a fixed local routing, attention computes its routing from the input itself, allowing content-dependent long-range relationships to be discovered.
- This dynamic, input-dependent computation is one of the architecture's deepest properties.

4.9 Attention Mechanism

- Introduces scaled dot-product attention as the differentiable building block of the transformer: queries, keys and values, each obtained by linear projection of the input, combine through a softmax over compatibility scores to produce weighted averages of values.
- The submodule interprets softmax as a differentiable soft lookup, justifies the $1/\sqrt{d}$ scaling, and reads attention as Nadaraya–Watson kernel regression with a learnable kernel.

4.10 Multi-head Attention

- Argues that a single attention pattern cannot simultaneously capture multiple overlapping relationships (syntactic, semantic, positional) and introduces multi-head attention as a partition of the representation into h parallel subspaces, each with its own projections.
- Empirical studies show heads specialising for positional adjacency, syntactic dependency and coreference without any explicit supervision — ensembling within a single layer.

4.11 Transformer Blocks

- Assembles attention, feed-forward, residual connections and layer normalisation into the repeated block that is the transformer's computational unit.
- The pre-norm formulation, now standard in modern transformers, is contrasted with post-norm in terms of gradient flow and training stability.

4.12 Positional Encoding

- Explains that attention is permutation-equivariant, so position information must be explicitly injected.
- The submodule catalogues the major schemes — sinusoidal, learned, rotary (RoPE) and ALiBi — and explains their distinct mathematical properties, with RoPE applying position multiplicatively as a rotation so that dot products depend only on relative offset.

4.13 Residual Connections

- Develops the residual stream interpretation: the representation at any depth is the sum of the original embedding and all updates applied by preceding layers.
- The residual stream creates implicit skip connections across the entire network, allowing information from early layers to remain accessible to later ones and enabling stable training at depth.

4.14 Feed Forward Networks

- Describes the per-position two-layer feed-forward network as the transformer's nonlinear capacity.
- With an inner dimension typically four times d_{model} and an activation like GELU or SwiGLU, the FFN is increasingly understood as an associative-memory component whose first layer's weights encode keys and second layer's weights encode the corresponding values.

4.15 Complete Transformer Pipeline

- Assembles the complete forward pass: tokens are embedded, positional information is added, the sequence flows through N transformer blocks with residual connections, a final layer normalisation is applied, and a task-specific head produces the output.
- The residual stream is read as a broadcast communication channel and depth as iterative refinement.

4.16 Why Transformers Work

- Offers four complementary explanations for the transformer's empirical dominance: universal approximation (necessary but insufficient), dynamic input-dependent computation, the elimination of the long-range gradient bottleneck that handicaps recurrent networks, and the quadratic computational cost that has motivated efficient-attention variants such as linear attention, sparse attention, sliding windows and FlashAttention.

4.17 Transformers Across Domains

- Demonstrates the transformer's domain-generalty by surveying applications: autoregressive language modelling with causal masking; Vision Transformers that treat images as sequences of patches and match convolutional architectures despite using none of their image-specific inductive biases; and multimodal transformers that process heterogeneous inputs in a single shared attention mechanism.

Module 5 — Optimisation: Discovering Useful Solutions

Theme *Learning is optimisation.*

Module 5 develops the search layer. Once an architecture defines a hypothesis space and a loss defines what success means, optimisation is the process by which good parameters are actually found. The module covers gradient descent and its modern variants, the geometry of high-dimensional loss landscapes, the family of regularisation methods, and the deep puzzle of why simple gradient-based methods navigate non-convex landscapes so successfully.

5.1 Language of Optimisation

- Introduces optimisation as the engine of learning: a loss function measures how wrong a model is, and training is the search for parameters that minimise it. The gradient points uphill, gradient descent steps in the opposite direction, and the learning rate controls step size.
- Classical convex optimisation theory provides strong guarantees that do not directly apply to neural networks, whose loss landscapes are non-convex yet usually navigable in practice.

5.2 Training Loop

- Walks through the canonical training loop for a neural network: sample a mini-batch, forward-propagate to compute predictions, evaluate the loss, back-propagate to compute gradients, take an optimiser step, repeat.
- The submodule frames the training loop as the operational realisation of every theoretical idea in the course.

5.3 Loss Functions

- Defines the loss as a non-negative scalar map measuring discrepancy between predictions and targets, and argues that the choice of loss is the mathematical specification of what learning means.
- The canonical losses are catalogued – MSE for regression (Gaussian-noise MLE), cross-entropy for classification (categorical MLE), and hinge loss for SVMs – with each loss shown to embed a philosophy about what 'good performance' means.

5.4 Loss Landscapes

- Treats the loss as a surface over parameter space and develops the geometric vocabulary needed to reason about it: critical points, gradient, Hessian.
- High-dimensional landscapes are qualitatively unlike low-dimensional ones – most critical points are saddle points, true local minima tend to lie close to the global minimum, and vast connected regions of near-equal loss exist.

5.5 Gradient Descent

- Develops gradient descent as the canonical optimiser: compute the gradient, step in the opposite direction, repeat.
- The submodule explains the role of the learning rate and distinguishes batch gradient descent (exact but expensive), stochastic gradient descent (noisy but cheap), and mini-batch SGD as the practical compromise that dominates modern deep learning.

5.6 Stochastic Gradient Descent

- Develops mini-batch SGD as the workhorse of modern training.
- The submodule explains the noise of stochastic gradients as a form of implicit regularisation, biasing trajectories toward flat minima that correlate with generalisation, and shows how batch size, learning rate and noise interact.

5.7 Momentum

- Introduces momentum as an exponentially weighted average of past gradients, which accelerates descent through narrow valleys and dampens oscillations.
- Nesterov accelerated gradient is described as a look-ahead variant that evaluates the gradient at the projected next position.

5.8 Adam

- Develops Adam as the combination of momentum (first-moment estimate of the gradient) and RMSProp (second-moment estimate of squared gradients) with bias correction for the early steps. AdamW is introduced as the modern default, decoupling weight decay from the adaptive gradient update for cleaner regularisation behaviour.

5.9 Learning Rates

- Argues that the learning rate is the single most consequential hyperparameter, controlling both convergence speed and oscillation radius near minima.
- Standard schedules are catalogued — warmup, step decay, cosine annealing, cyclical and one-cycle policies — with cosine annealing identified as the modern default for large language models and warmup as essential for transformer training.

5.10 Regularisation

- Introduces regularisation as a principled response to overfitting: penalise large parameter values to trade a small increase in bias for a substantial reduction in variance.
- The submodule surveys the family — L1, L2, elastic net, dropout, weight decay, data augmentation — and frames each as encoding a prior belief about what plausible solutions look like.

5.11 Ridge Regression

- Develops L2 regularisation through the ridge estimator $(X^T X + \lambda I)^{-1} X^T y$, explaining why the added λI guarantees invertibility and reading ridge geometrically as constraining the solution to an L2 ball whose radius shrinks with λ .

5.12 Lasso

- Replaces the L2 penalty with an L1 penalty and explains the geometric consequence: the constraint region becomes a diamond whose corners lie on coordinate axes, so the optimal solution tends to lie at a corner where some coordinates are exactly zero.
- Lasso therefore performs automatic feature selection alongside estimation.

5.13 Elastic Net

- Combines L1 and L2 penalties to keep the sparsity-inducing behaviour of Lasso while restoring the stability of ridge when predictors are correlated.
- A mixing parameter interpolates between pure Lasso and pure ridge, with the submodule explaining when each extreme is appropriate.

5.14 Bias–Variance Trade-off

- Reads regularisation through the bias–variance lens: λ controls a trade-off between increased bias and reduced variance, and the optimal λ minimises total mean squared error.
- Cross-validation is introduced as the standard tool for finding it, and the classical U-curve in test error is shown as the visual signature of the trade-off.

5.15 Bayesian Interpretation of Regularisation

- Provides the elegant Bayesian reading: regularised regression is maximum-a-posteriori estimation with a specific prior.
- Ridge corresponds to a Gaussian prior, Lasso to a Laplace prior, Elastic Net to a mixed prior. Choosing a penalty is choosing a prior belief, and this view unifies regularisation with the broader Bayesian framework.

5.16 Lottery Ticket Hypothesis

- Introduces Frankle and Carbin's hypothesis that a randomly initialised dense network contains a sparse subnetwork — a winning ticket — that, when trained in isolation from its original initialisation, reaches comparable test accuracy.
- The over-parameterised network is reframed as a search space whose vast capacity makes it overwhelmingly likely that a good subnetwork exists at initialisation, and training is the process of identifying it.

5.17 Optimisation in Deep Learning

- Closes the module by explaining why deep learning routinely ignores the classical prescription to regularise heavily. Implicit regularisation comes from multiple sources — architectural constraints, SGD's gradient noise biasing trajectories toward flat minima, and gradient descent's preference for minimum-norm interpolating solutions — providing a principled guide to when explicit regularisation still pays.

Module 6 — Theory: Why AI Works

Theme *Explaining the surprising success of deep learning.*

Module 6 develops the explanatory layer. Classical statistical learning theory predicted that modern over-parameterised networks should fail, yet they succeed spectacularly. The module surveys the most useful partial explanations available — implicit regularisation, the manifold hypothesis, double descent, benign overfitting, scaling laws, mechanistic interpretability — and is honest that the destination is a frontier rather than a closed proof.

6.1 Classical Statistical Learning Theory

Reviews the classical apparatus — bias–variance, VC dimension, Rademacher complexity, uniform convergence, structural risk minimisation — that dominated statistical learning theory for half a century. These frameworks remain mathematically correct within their stated assumptions but were developed for a regime quite different from the high-dimensional, gradient-descent-trained systems used today.

6.2 Why Overparameterisation Should Fail

States the classical warning clearly: when the number of parameters p vastly exceeds the number of training examples n , VC dimension and Rademacher complexity predict catastrophic overfitting. The submodule traces why practitioners pushed past this warning anyway and frames the resulting puzzle as the defining empirical mystery of modern machine learning.

6.3 Double Descent

Documents the phenomenon, due to Belkin, Nakkiran and colleagues, that as model complexity is pushed well beyond the interpolation threshold, test error decreases again, sometimes below the classical optimum. The full curve has three regions — the classical regime, the spike near interpolation, and the over-parameterised regime — and modern deep learning lives in the third.

6.4 Benign Overfitting

Examines the surprising finding that modern networks can perfectly fit (even noisy) training data and still generalise well — overfitting that is, in a precise mathematical sense, benign. The submodule connects benign overfitting to the manifold hypothesis and to the implicit bias of gradient descent toward minimum-norm interpolating solutions.

6.5 Manifold Hypothesis

Revisits the manifold hypothesis from a theoretical angle. The effective dimensionality of the learning problem is governed by the manifold's intrinsic dimension rather than the ambient dimension, and excess parameters become the resource the model uses to chart the manifold's geometry at multiple scales — turning what classical theory saw as wasted capacity into the substrate that makes generalisation possible.

6.6 Scaling Laws

Develops the empirical scaling laws of Kaplan and colleagues: cross-entropy loss decreases as a clean power law in model parameters, training tokens and compute budget. The Chinchilla correction is then presented, showing that optimal compute allocation splits resources roughly equally between model size and dataset size — a finding that retroactively reframed pre-2022 large language models as significantly under-trained.

6.7 Emergent Abilities

Examines the phenomenon, catalogued by Wei and colleagues, that certain abilities – multi-step arithmetic, chain-of-thought reasoning, in-context learning – appear abruptly at specific model scales rather than improving smoothly. Scaling laws are predictive of loss but explicitly silent on which capabilities a given loss level will unlock.

6.8 Lottery Ticket Hypothesis Revisited

Returns to the lottery ticket hypothesis with the stronger version (subnetworks at initialisation that already perform well) and recent theoretical results connecting the phenomenon to implicit bias, neural tangent kernel theory and the structure of high-dimensional loss landscapes.

6.9 Mechanistic Interpretability

Treats mechanistic interpretability as theory rather than tooling: the reverse-engineering of internal circuits provides empirical evidence about what trained networks actually compute, constraining and informing the theory of why they generalise. Induction heads, indirect object identification and similar discoveries are introduced as exemplars of the programme.

6.10 Compression as Intelligence

Develops the information-theoretic view of learning as compression: a good model is one that gives the data a short description, and compression is therefore a proxy for understanding. The submodule connects this view to PAC-Bayes bounds and to the broader theoretical link between description length and generalisation.

6.11 Information Theory and Learning

Develops Shannon's information theory as the theoretical foundation for loss functions, model comparison and the concept of learning itself. Entropy measures average surprise; cross-entropy measures the inefficiency of encoding samples from one distribution under an encoding optimised for another; KL divergence measures distributional difference. Minimising cross-entropy in training is exactly minimising KL between the model's predictions and the empirical label distribution.

6.12 Open Problems in AI Theory

Catalogues the most important unanswered questions: why deep learning generalises (still without a unified account), what the correct complexity measure for neural networks is (parameter count is clearly wrong), whether emergent capabilities can be predicted before they appear, and where current understanding breaks down. The submodule offers these as invitations to engage with open problems.

Module 7 – Mathematical Models Across AI

Theme *The mathematical evolution of machine learning.*

Module 7 traces the unbroken arc from linear regression to modern generative AI. Each model in the sequence adds one mathematical ingredient – nonlinear features, nonlinear outputs, learned representations, compositional depth, dynamic input-dependent selection – and the arc from $y = w^T x + b$ to GPT is shown to be, mathematically, unbroken.

7.1 Linear Models

Recovers a precise meaning for the loosely-used word 'linear': a model is linear in its parameters, not necessarily in its inputs. The submodule traces why linearity dominated statistical modelling for two centuries – closed-form least-squares solutions, a complete inference apparatus, and computational tractability – and lays out the three fundamental limitations the rest of the module progressively overcomes.

7.2 Linear Regression

Develops linear regression as the concrete realisation of the linearity assumption: the OLS estimator $\beta = (X^T X)^{-1} X^T y$, its geometric interpretation as projection onto the column space, the classical statistical assumptions, the Gauss–Markov optimality theorem, and the interpretation of multiple-regression coefficients as partial effects. The conditions under which linear regression fails are catalogued.

7.3 Polynomial Regression

Extends the linear framework by transforming inputs through pre-specified basis functions while keeping parameters linear. Polynomial, trigonometric, Gaussian radial, B-spline and wavelet expansions all fit within the same OLS machinery applied to a richer design matrix. Runge's phenomenon – high-degree polynomials oscillating wildly between data points – motivates the need for regularisation.

7.4 Generalised Linear Models

Presents Nelder and Wedderburn's GLM framework – random component (exponential-family distribution), systematic component (linear predictor), link function – as the bridge between classical statistics and machine learning. Link functions anticipate activation functions, IRLS anticipates gradient descent, and a logistic regression unit is precisely a single neuron.

7.5 Logistic Regression

Identifies binary logistic regression as a Bernoulli GLM with the logit link, and points out that a single sigmoid neuron is exactly a logistic regression unit. The submodule develops the maximum-likelihood objective, the IRLS algorithm and the deviance-based comparisons that connect this classical method to modern neural networks.

7.6 Support Vector Machines

Develops the SVM as the apex of the linear paradigm: a maximum-margin classifier whose dual formulation depends on data only through inner products, enabling the kernel trick that extends linear classifiers to nonlinear decision boundaries via implicit high-dimensional feature spaces. The submodule covers hard- and soft-margin formulations, the dual problem, the kernel catalogue, and why SVMs ceded ground when deep networks began to learn their own representations.

7.7 Multi-layer Perceptrons

Introduces the MLP as the architecture that finally breaks the linear paradigm by composing affine maps with nonlinear activations across multiple layers. The submodule covers the role of nonlinearity, the emergence of learned representations, the vanishing and exploding gradient problems, the chain rule and matrix representation of backpropagation, and the central observation that backpropagation's contribution is computational efficiency.

7.8 Unified Function Approximation

Closes the linear-to-nonlinear progression by placing every model – linear regression, polynomial, GLM, SVM with kernels, shallow and deep MLPs, transformers – in a common hypothesis-class framework. What changes across the arc is the inductive bias: which functions are easy to represent and how the optimisation landscape shapes the solutions actually found.

7.9 Generative Models

Contrasts discriminative modelling (learning $p(y|x)$) with generative modelling (learning $p(x)$) and distinguishes three increasingly difficult capabilities: density evaluation, sampling, and both together. Generative modelling is fundamentally harder because it must account for the entire structure of the input space, and its history is read as a sequence of clever strategies for evading the curse of dimensionality.

7.10 Autoencoders

Introduces the encoder–decoder framework as the simplest architecture that embodies the generative idea. An encoder compresses input to a low-dimensional latent code, and a decoder reconstructs the input from it; reconstruction loss forces the bottleneck to retain useful structure. The bottleneck recovers PCA in the linear case and richer features in the deep case.

7.11 Variational Autoencoders

Develops Kingma and Welling's VAE as the conceptual shift from compression to generation. Instead of mapping each input to a single latent point, the VAE maps it to a distribution, forcing the latent space to be smooth and samplable. Training balances reconstruction accuracy against a KL penalty that aligns the encoder distribution with a simple prior.

7.12 GANs

Describes Goodfellow's GAN as a game between a generator that transforms noise into samples and a discriminator that distinguishes real from generated data. The two networks are trained simultaneously in a minimax game. The submodule explains why GANs produce sharper images than VAEs and catalogues the well-known difficulties — instability, mode collapse, hyperparameter sensitivity — that motivated decades of subsequent work.

7.13 Diffusion Models

Presents diffusion models as a different class of generative system that gradually destroys data with Gaussian noise in a forward process and learns to reverse the noising step by step. The submodule walks through the forward equation, the simple mean-squared-error training objective on predicted noise, and the iterative sampling procedure, reading diffusion as iterative refinement: structure emerging from randomness through a sequence of small, learnable corrections.

7.14 Mathematics Unifying Generative AI

Identifies the shared mathematical structure beneath VAEs, GANs and diffusion models: every generative model learns a transport map that pushes samples from a simple base distribution to the data distribution. The models differ in how they parameterise and train this map — variational inference (VAE), adversarial transport (GAN) or iterative transport (diffusion).

7.15 Neural versus Non-neural Models

Surveys the structural advantages of non-neural approaches — kernel methods, Gaussian processes, tree ensembles, Bayesian networks — and argues for studying them on four grounds: theoretical guarantees, practical superiority on specific problem classes (tabular data), interpretability and verifiability, and data and computational efficiency where neural networks are disqualifying.

Module 8 — Frontiers of Artificial Intelligence

Theme *Current research and open questions.*

8.1 Mechanistic Interpretability

- Treats mechanistic interpretability as a research programme rather than a finished discipline: reverse-engineering neural networks at the level of internal computations, with circuits, induction heads, indirect object identification and superposition as exemplars.
- The frontier is the move from individual case studies to a general theory of how trained networks organise their computation.

8.2 Scaling Laws

- Surveys scaling as an active research area: the Kaplan and Chinchilla laws, their successors, the role of data quality, and the open question of whether scale alone is sufficient.
- The submodule examines what scaling laws predict, what they cannot predict (emergence), and where the empirical curves are likely to break.

8.3 World Models

- Introduces the research programme of building learned models of environment dynamics that support planning, simulation and counterfactual reasoning.
- World models are framed as the integration of representation learning with structured prediction, with applications spanning reinforcement learning, robotics and embodied AI.

8.4 Scientific Machine Learning

- Surveys the application of machine learning to scientific discovery: physics-informed neural networks, neural differential equations, learned surrogates for expensive simulations, and the discovery of new materials, molecules and proteins.
- AlphaFold is treated as the exemplar of the broader programme.

8.5 Symbolic AI

- Surveys the first four decades of AI, when symbolic methods — logical propositions, production rules, semantic networks, frames, ontologies — dominated, and contrasts these with current statistical approaches.
- The submodule extends to the neuro-symbolic synthesis that seeks to combine the pattern recognition of neural networks with the reasoning and verifiability of symbolic systems.

8.6 Tree Ensembles

- Explains why gradient-boosted decision tree ensembles continue to dominate on tabular data despite the rise of deep learning.
- Tabular data consists of semantically heterogeneous features that neural networks struggle with because they must embed all features into a common vector space and learn interactions through smooth nonlinearities, whereas tree splits handle heterogeneity natively.

8.7 Foundation Models

- Frames foundation models as the logical endpoint of transfer learning: a single, broadly-trained representation that amortises the cost of learning general structure across all downstream tasks.
- The submodule examines the pre-training objectives, the scaling regime, and the open question of how foundation models acquire what look like reasoning capabilities.

8.8 Multimodal AI

- Surveys the move from single-modality to cross-modality models. Tokenising vision, audio, language and other modalities into a shared representation space allows a single transformer to process all of them, and the alignment that emerges between modalities is read as evidence for the platonic representation hypothesis.

8.9 Agentic AI

- Introduces agentic AI as the integration of foundation models with tools, memory and environment interaction. Agents plan, act, observe and replan, raising research questions in long-horizon credit assignment, tool use, multi-agent coordination, and the safety and governance of systems that act in the world.

8.10 AI Beyond Neural Networks

- Argues that no single class of model dominates universally. Tree ensembles dominate tabular data, kernel methods retain theoretical guarantees, Gaussian processes provide calibrated uncertainty, and evolutionary methods solve non-differentiable problems.
- The No Free Lunch theorem implies that what matters is the match between an algorithm's inductive biases and the structure of the problem.

8.11 Open Research Questions

- Catalogues the open questions that animate current research: why deep learning generalises, what the correct complexity measure for neural networks is, whether emergent capabilities can be predicted before they appear, how to align increasingly capable systems with human intent, and what comes after the transformer.
- The submodule offers these as invitations rather than as a closing summary.

Process, Evaluations and Hall of Fame

Process

- Self paced
- Buddy system if desired
- Typically months but not mandatory
- Questions will be answered online within the week
- We will have office hours sessions where you can ask questions live. Office hours sessions are designated, recurring periods of time where experts, professors, managers, or mentors make themselves available to offer personalized guidance, answer questions, and facilitate collaboration

Process for evaluations

- Scenario driven evaluation (example below)
- You can use AI for assistance
- Scenarios based on mathematical concepts / toolkit
- There is no specific way to address scenario but your thinking process is key
- 25 scenarios to complete course.

- We will also have a framework to apply this analysis to research papers. That will get you into an (optional) hall of fame
- We can also work on a buddy system and the chat/forum (optional)

Mathematical concepts/ Toolkit used for scenarios

Statistical learning

1. Bias–Variance Tradeoff
2. Overfitting vs Underfitting
3. Generalisation
4. Inductive Bias
5. Empirical Risk Minimisation (ERM)
6. Structural Risk Minimisation (SRM)
7. VC Dimension
8. Probably Approximately Correct (PAC) Learning
9. No Free Lunch Theorem
10. Double Descent

Probability and uncertainty

11. Bayes' Theorem
12. Bayesian Updating
13. Maximum Likelihood Estimation (MLE)
14. Maximum A Posteriori (MAP)
15. Prior and Posterior Distributions
16. Conditional Probability
17. Markov Property
18. Hidden Markov Models
19. Gaussian Distributions
20. Monte Carlo Sampling

Information theory

21. Entropy
22. Cross Entropy
23. KL Divergence
24. Mutual Information
25. Information Gain
26. Minimum Description Length (MDL)
27. Compression as Intelligence
28. Kolmogorov Complexity
29. Fisher Information
30. Rate–Distortion Theory

Optimisation

31. Gradient Descent
32. Stochastic Gradient Descent
33. Convex Optimisation
34. Non-convex Optimisation
35. Loss Functions
36. Regularisation (L1/L2)
37. Lagrange Multipliers

- 38. Saddle Points
- 39. Newton's Method
- 40. Momentum Optimisation

Representation learning

- 41. Feature Engineering
- 42. Latent Representations
- 43. Embedding Spaces
- 44. Dimensionality Reduction
- 45. Principal Component Analysis (PCA)
- 46. Manifold Learning
- 47. Kernel Methods
- 48. Sparse Representations
- 49. Autoencoders
- 50. Representation Learning

AI theories

- The Lottery Ticket Hypothesis
- Leo Breiman's Two Cultures of Statistical Modeling
- Scaling Laws
- Neural Tangent Kernel
- Information Bottleneck Theory
- Free Energy Principle
- World Models
- Predictive Processing
- Causal Inference
- Counterfactual Reasoning
- Reinforcement Learning as Sequential Decision Making
- Sequential Decision Analytics
- Attention as Differentiable Memory
- Sparse Mixture of Experts
- Chain-of-Thought Reasoning
- Test-Time Compute
- In-Context Learning
- Retrieval-Augmented Generation (RAG)
- Agentic Planning

Example of a Scenario driven Quiz

Instead of asking: "What is entropy?" We present students with a realistic scenario:

A hospital wants to predict patient readmission.

A model achieves 99% accuracy but almost never detects patients who will actually be readmitted.

Using the five mathematical lenses:

- What learning problem is being solved?
- Which mathematical concept is most relevant?
- How is it computed?
- What quantity is being optimised?
- Which mathematical theory explains why the current model fails?

The answer might revolve around **cross-entropy**, **class imbalance**, **information gain**, **precision/recall**, and **generalisation**, requiring students to connect mathematics to AI rather than recall definitions. There is also no one specific answer in this case.

Please contact me if you want to join the course

Mathematical Foundations of Artificial Intelligence - *A Universal Framework for Understanding Modern AI* By [Ajit Jaokar](#) ajit.jaokar [at] gmail [dot] com